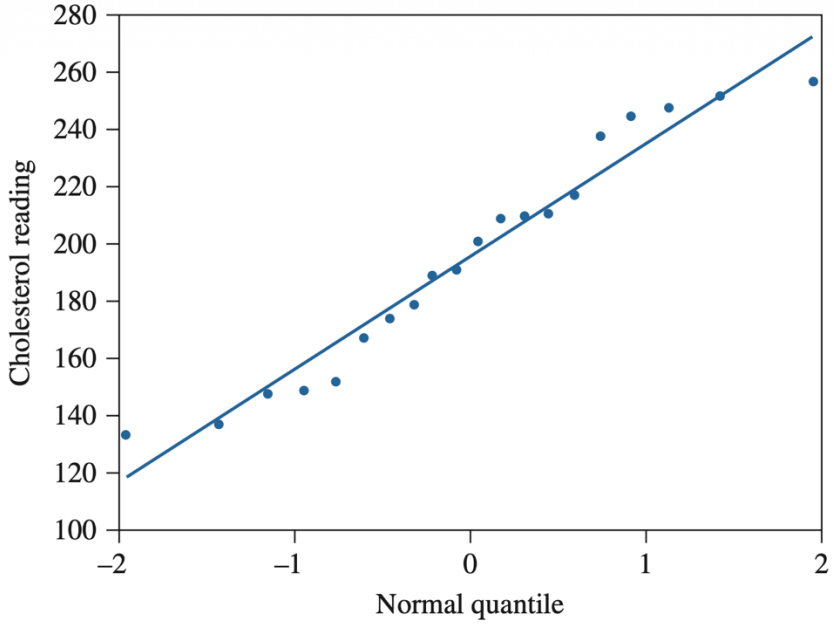
FIGURE 4.28
Normal quantile plot for cholesterol reading



4.16 R Instructions

Generating Random Numbers

To generate 1,000 random numbers from the integers [0, 1,…, 9]:

1. $y = c(0{:}9)$
2. x = sample(y, 1000, replace=T)
3. x

Calculating Binomial Probabilities

To calculate binomial probabilities when X has a binomial distribution with $n = 10$ and $\pi = 0.6$:

1. To calculate $P(X = 3)$, use the command **dbinom(3, 10, .6)**
2. To calculate $P(X \leq 3)$, use the command **pbinom(3, 10, .6)**
3. To calculate $P(X = k)$ for $k = 0, 1, \ldots, 10$, use the commands k = c(0 : 10) and **dbinom(*k*, 10, .6)**

Calculating Poisson Probabilities

To calculate Poisson probabilities when Y has a binomial distribution with $\lambda = 10$ and $\pi = 0.6$:

1. To calculate $P(X = 3)$, use the command **dbinom(3, 10, .6)**
2. To calculate $P(X \leq 3)$, use the command **pbinom(3, 10, .6)**
3. To calculate $P(X = k)$ for $k = 0, 1, \ldots, 10$, use the commands k = c(0 : 10) and **dbinom(*k*, 10, .6)**

Calculating Normal Probabilities

To calculate probabilities when X has a normal distribution with $\mu = 23$ and $\sigma = 5$:

1. To calculate $P(X \leq 18)$, use the command **pnorm(18, 23, 5)**
2. To calculate $P(X > 18)$, use the command **1 − pnorm(18, 23, 5)**
3. To find 85th percentile, use **q(.85, 23, 5)**

Generating Sampling Distribution of $\bar{y}$

The following R commands will simulate the sampling distribution of $\bar{y}$. We will generate 10,000 values of $\bar{y}$, with each of the 10,000 values of $\bar{y}$ computed from a unique random sample of 16 observations, from a population having a normal distribution with $\mu = 43$ and $\sigma = 7$.

1. $r = 10{,}000$
2. y = rep(0, 16)
3. ybar16 = rep(0, r)
4. for (i in 1:r){
5. y = ***r*norm**(16, 43, 7)
6. ybar16[i] = **mean**(y) }

The above commands will produce 10,000 values for $\bar{y}$, where $\bar{y}$ is the average of 16 data values from a population having a normal distribution with $\mu = 43$ and $\sigma = 7$. To display the 10,000 values, type "ybar16".

The following three commands will generate a histogram, mean, and standard deviation for the 10,000 values:

1. **hist**(ybar16)
2. **mean**(ybar16)
3. **sd**(ybar16)

The histogram should be bell-shaped with its center near 43. The mean of the 10,000 values should be close to 43, and the standard deviation should be close to $7/\sqrt{16} = 1.75$.

Commands to Generate the Plot in Figure 4.28

The following R commands will generate the normal reference plot in Figure 4.28 and the correlation coefficient.

1. $y = c$(133, 137, 148, 149, 152, 167, 174, 179, 189, 192, 201, 209, 210, 211, 218, 238, 245, 248, 253, 257)
2. y = sort(y)
3. n = length(y)
4. i = 1 : n
5. $u = (i - .375)/(\text{n} + .25)$
6. x = **qnorm**(u)
7. plot(x, y, xlab = "Normal quantiles", ylab = "Cholesterol readings", lab = c(7, 8, 7), ylim = c(100, 280), main = "Normal Reference Distribution Plot\n Cholesterol readings", cex = .95)
8. **abline**(lm(y ~ x))
9. **cor**(x, y)